

Active arts: analysis

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K^* \rho)_\tau f) \\ f(t=0) &= f_0 \end{cases}$$

(dV, Bruna & Burger, SIAM J. Appl. Dyn. Syst., 2025):

- PDE always globally-in-time well-posed for singular Newtonian ∇K and $\sigma_x \geq 0$

◦ For any $T \geq 0$ $f \in L^2(0, T; H^1) \cap L^\infty(0, T; L^2)$

• **No blow-up in infinite time (Alikos, CPDE, 1979)**

◦ For $f_0 \in L^\infty \cap L^1$: $\sup_{t \geq 0} \|f(t)\|_{L^\infty} < C$

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(dW, Bruna & Burger, SIAM J. Appl. Dyn. Syst., 2025):

- PDE always globally-in-time well-posed for singular Newtonian ∇K and $\sigma_x > 0$
 - For any $T \geq 0$ $f \in L^2(0, T; H^1) \cap L^\infty(0, T; L^2)$
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 - For $f_0 \in L^\infty \cap L^1$: $\sup_{t \geq 0} \|f(t)\|_{L^\infty} < C$

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